

# **DESIGN and FABRICATION PROJECT**

## **(DS3014)**

**Automatic smart solutions of water purifier and  
classification system for different nature resources  
of water.**

### Team Members

|                     |            |
|---------------------|------------|
| Amit Kumar          | : 23BSM008 |
| Rahul Yadav         | : 23BEC089 |
| Banothu Ganesh      | : 23BCS055 |
| Prince Kumar Gautam | : 23BCS202 |
| Prince Raj Verma    | : 23BCS204 |
| Saurav Ratan        | : 23BCS227 |

DR. Avinash Ravi Raja

## **Problem Description :**

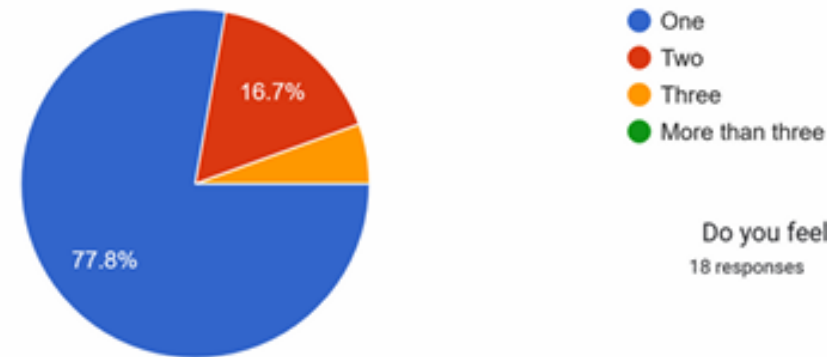
Water from natural sources like rain, river, pond, and sea contains different types of impurities such as dust, salts, and microorganisms. Normally, to purify each type of water, separate machines or systems are required because every source needs a different process.

- To solve the problem, we are developing a Microprocessor-based Automatic Smart Water Purification and Classification System.
- Based on the detected Resource, it will automatically choose the right purification process, like distillation for ocean water or sedimentation for pond/river water and end Processes are same for all Resources of Water.
- One single machine will handle multiple water sources without any manual adjustment. This makes the purification automatic, efficient, and cost-effective.

# User Study

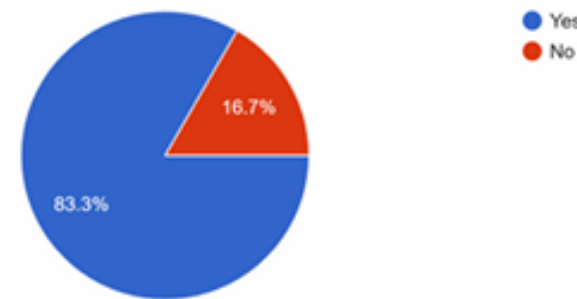
At your home, how many types of water sources does your current filtration system purify?

18 responses



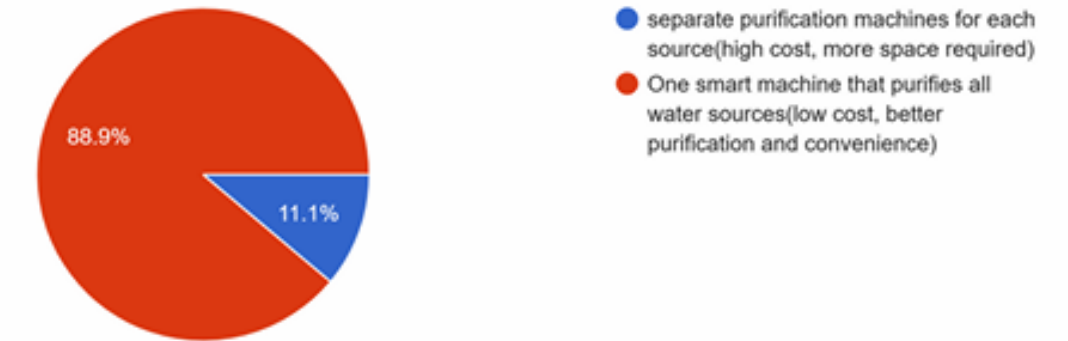
Do you feel purification requirements change based on water source?

18 responses



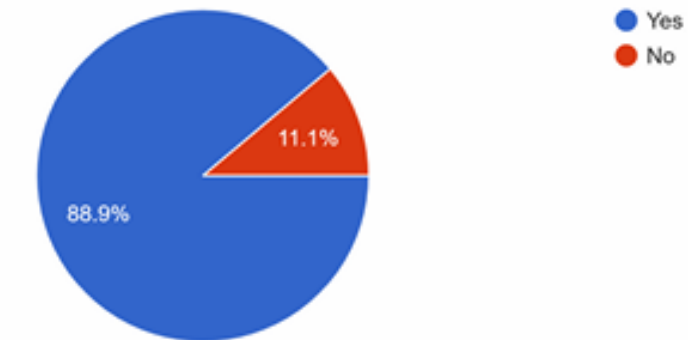
If you need to purify water from 2 or more sources (like rain, river, or ocean), which option would you prefer?

18 responses



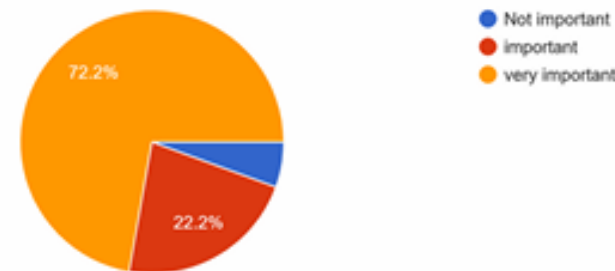
Would you use a device that automatically identifies the water source and selects purification steps?

18 responses



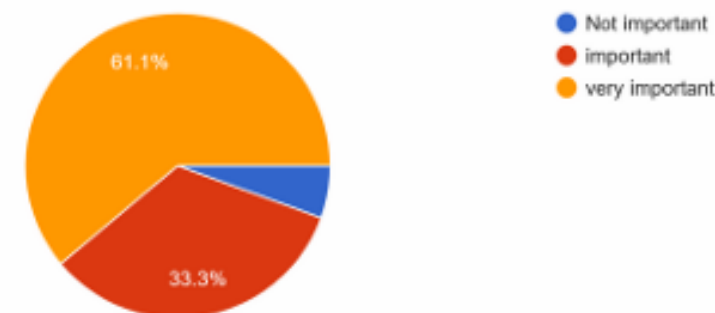
How useful is real-time pH & TDS monitoring for you?

18 responses



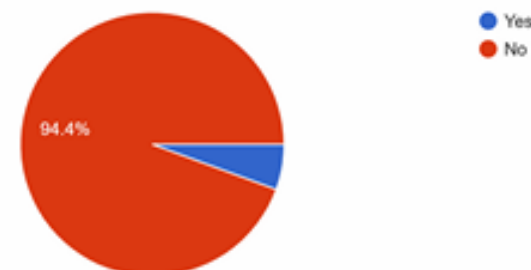
How important is it to have ONE system that purifies all types of water (rain, river, pond, ocean)?

18 responses

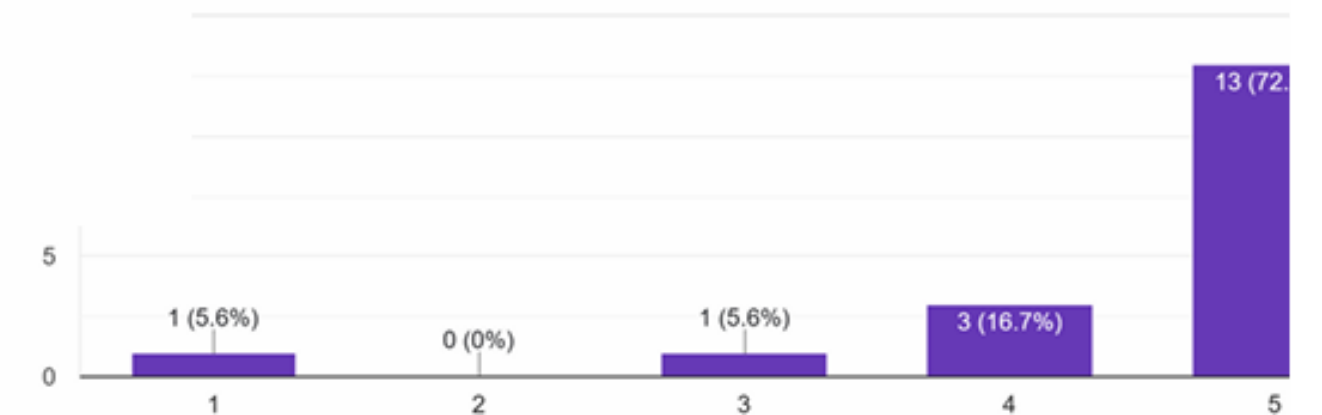


Does any purifier or machine in your home purify ocean/sea water and identify its salt type (like Black Sea, White Sea, or Red Sea)?

18 responses



can perform all these functions, how would you rate it out of 5?



# Water Quality Parameters :

| Water Source    | pH        | TDS (ppm)       | Turbidity (NTU) |
|-----------------|-----------|-----------------|-----------------|
| Rain Water      | 5.0 – 6.0 | 1 – 20          | 0 – 5           |
| Ground Water    | 6.5 – 8.5 | 200 – 2000      | 1 – 10          |
| River Water     | 6.5 – 7.5 | 50 – 250        | 10 – 100        |
| Pond Water      | 6.5 – 8.0 | 100 – 500       | 50 – 300        |
| Sea Water       | 7.5 – 8.4 | 35,000 – 45,000 | 1 – 20          |
| Ocean Water     | 7.6 – 8.3 | 33,000 – 38,000 | 0 – 10          |
| Drinkable Water | 6.5 – 8.5 | 50-150          | <1              |

# Impurities in Different Type of water resources:

## 1. Ground Water (Borewell water)

### a) Physical Impurities

- Suspended solids (silt, sand)
- Turbidity

### b) Chemical Impurities

- Hardness ( $\text{Ca}^{2+}$ ,  $\text{Mg}^{2+}$ )
- Iron ( $\text{Fe}^{2+}$ ), Manganese
- Fluoride
- Arsenic (in some regions)
- Nitrates and Nitrites
- Dissolved salts (TDS medium to high)

### c) Biological Impurities

- Usually low, but may contain:
- Bacteria (coliform)
- Protozoa

### d) Organic Impurities

- Very low; sometimes decayed underground matter

## 2. Rain Water

### a) Physical Impurities

- Dust particles from the atmosphere
- Micro suspended particles

### b) Chemical Impurities

- Dissolved gases ( $\text{CO}_2$ ,  $\text{SO}_2$ ,  $\text{NO}_2 \rightarrow$  acid rain)
- Low mineral content (soft water)

### c) Biological Impurities

- Airborne microbes
- Algae spores

### d) Organic Impurities

- Organic aerosols

## 3. River Water

### a) Physical Impurities

- Silt, clay, sand
- Mud (high turbidity)

### b) Chemical Impurities

- Moderate salts (TDS low)
- Industrial pollutants (in urban areas)
- Fertilizer runoff (nitrates, phosphates)
- Heavy metals in some regions

### c) Biological Impurities

- High:
  - Bacteria (coliform)
  - Viruses
  - Protozoa
  - Algae

### d) Organic Impurities

- Leaves, plant waste
- Organic pollutants (detergents, sewage)

## 4. Pond Water

### a) Physical Impurities

- Suspended solids (very high)
- Mud and decayed materials

### b) Chemical Impurities

- Fertilizer runoff
- High nutrient content (phosphates, nitrates)
- Low dissolved oxygen

### c) Biological Impurities

- Very high:
  - Algae (due to eutrophication)
  - Microorganisms
  - Insects, larvae

### d) Organic Impurities

- Decaying plants
- Animal waste
- Sewage mixing in rural ponds

## 5. Sea Water

### a) Physical Impurities

- Suspended solids (sand)
- Micro-sediments

### b) Chemical Impurities

- Very high TDS (32,000–40,000 mg/L)
  - NaCl (sodium chloride)
  - Magnesium, sulfate, calcium, potassium ions

- Industrial wastes near coastal regions

### c) Biological Impurities

- Marine microorganisms
- Algae
- Phytoplankton

### d) Organic Impurities

- Organic debris
- Oil pollution

## 6. Ocean Water

- Very similar to sea water, but deeper regions vary.

### a) Physical Impurities

- Suspended sediments
- Microplastic particles (increasing globally)

### b) Chemical Impurities

- High TDS (salts)
- Minerals and dissolved gases
- Trace metals

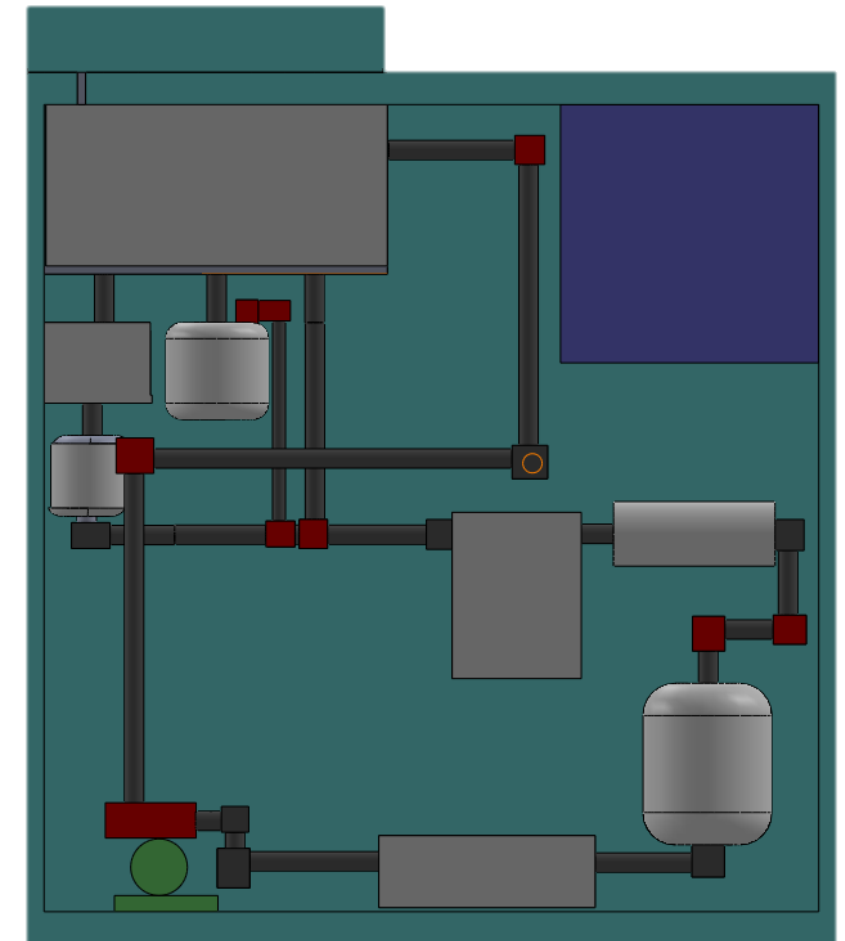
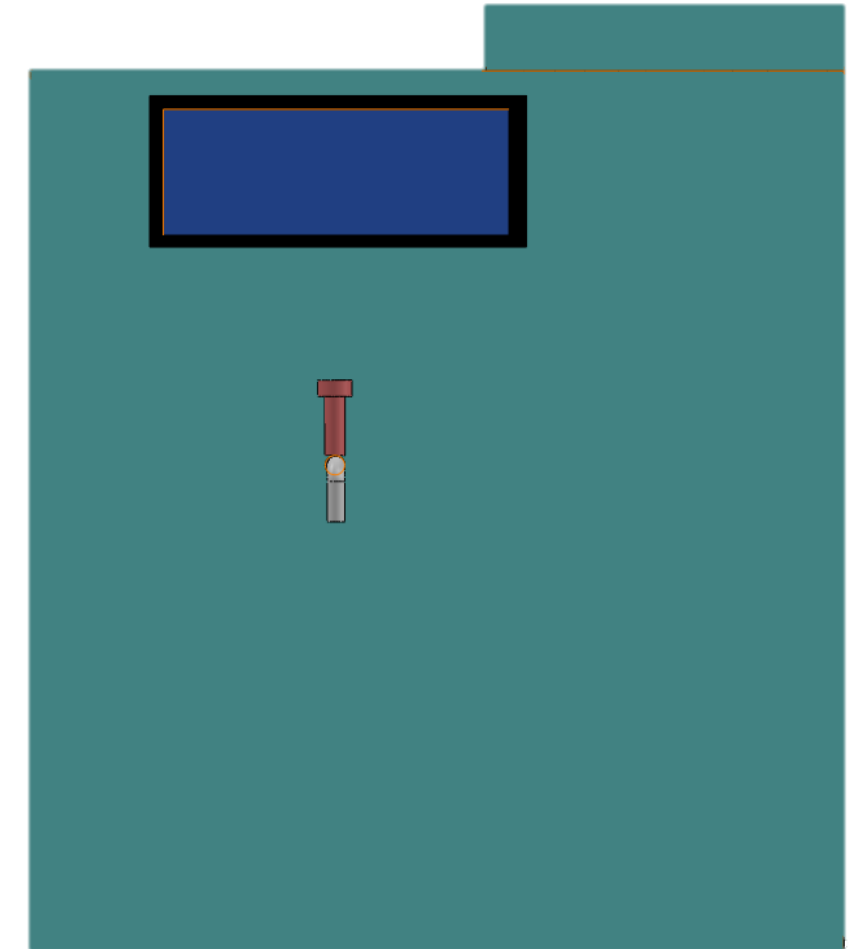
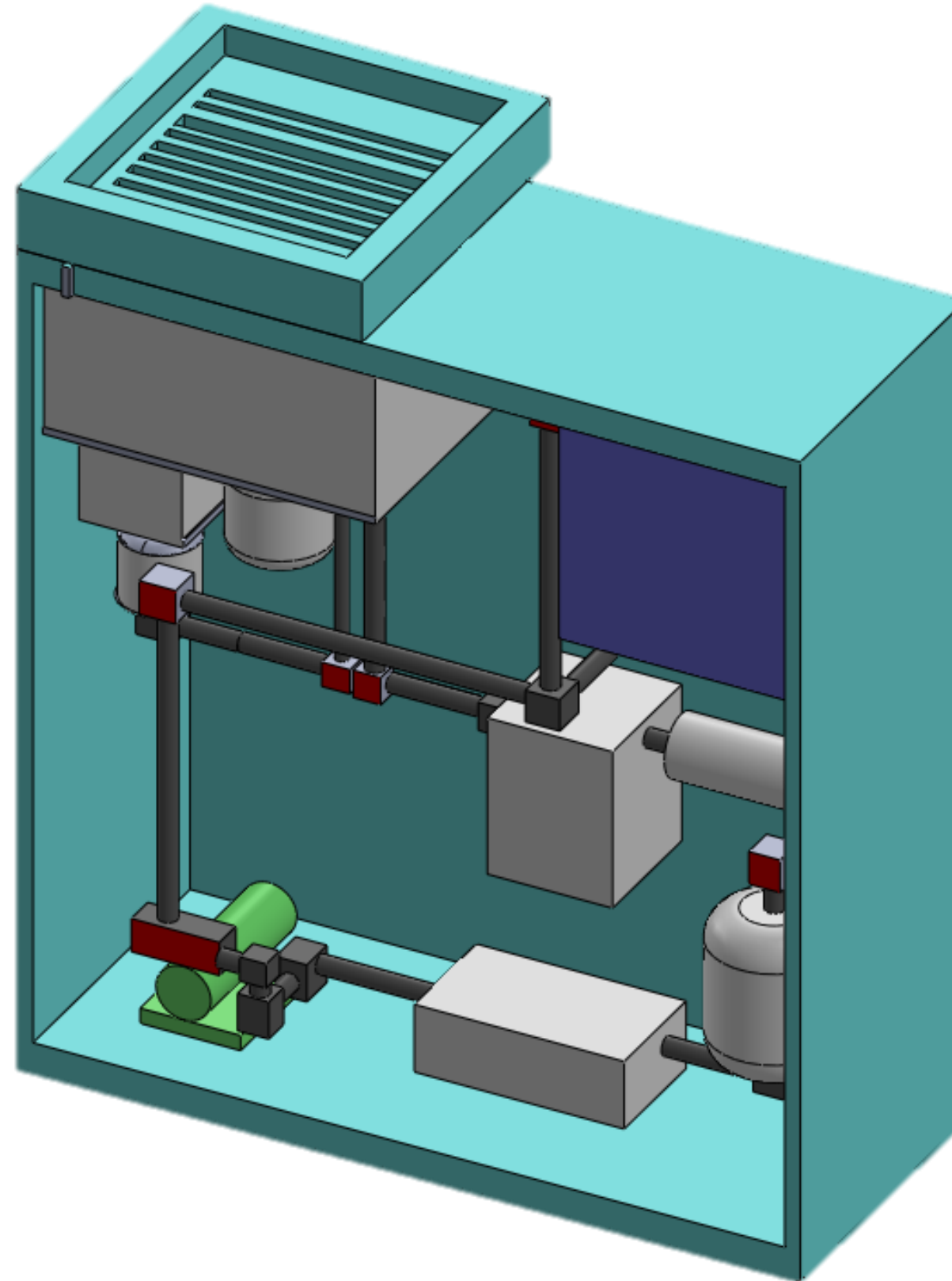
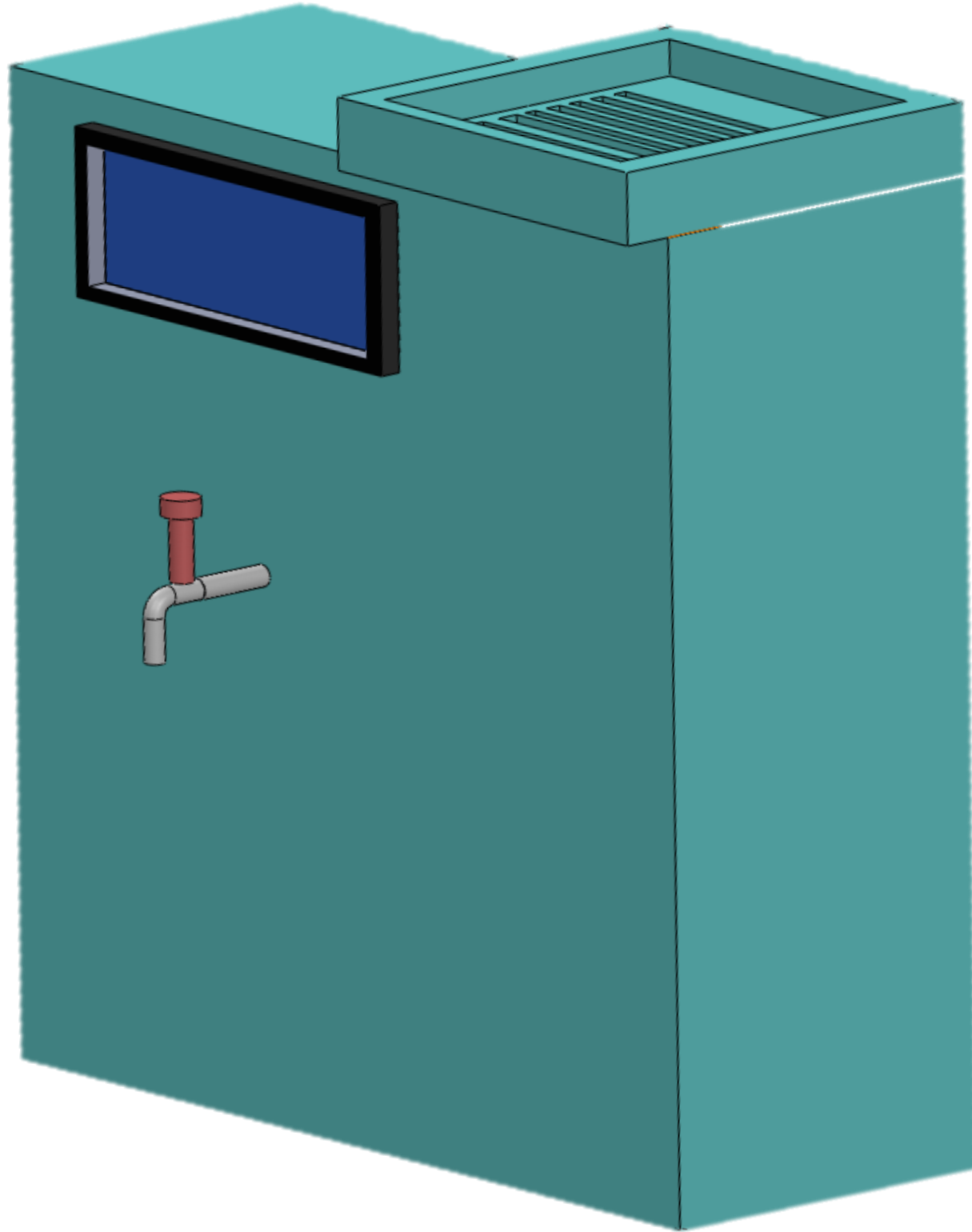
### c) Biological Impurities

- Marine microbes
- Plankton
- Organic particles

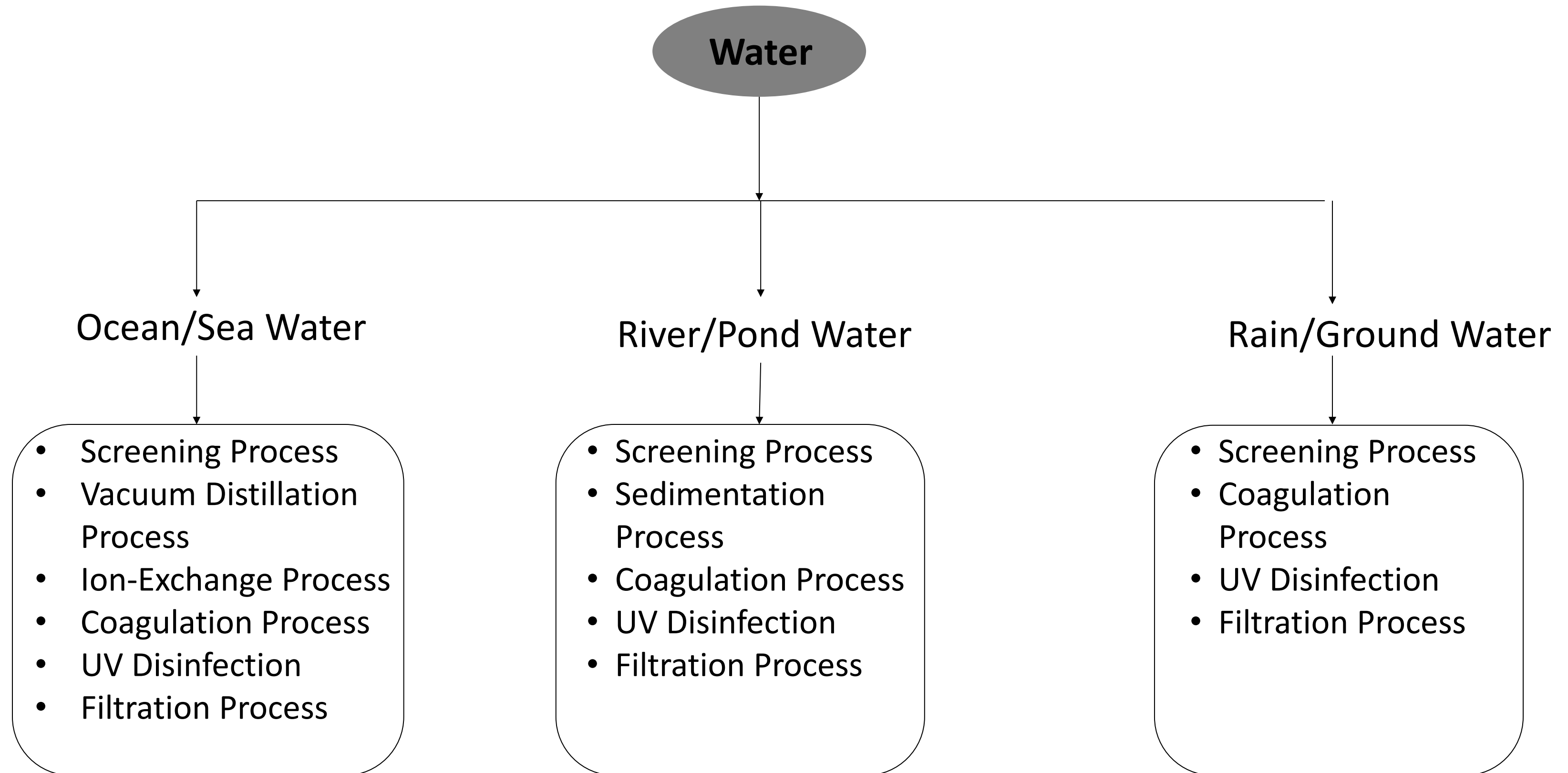
### d) Organic Impurities

- Dead organic matter
- Oceanic microplastics
- Oil traces (in industrial zones)

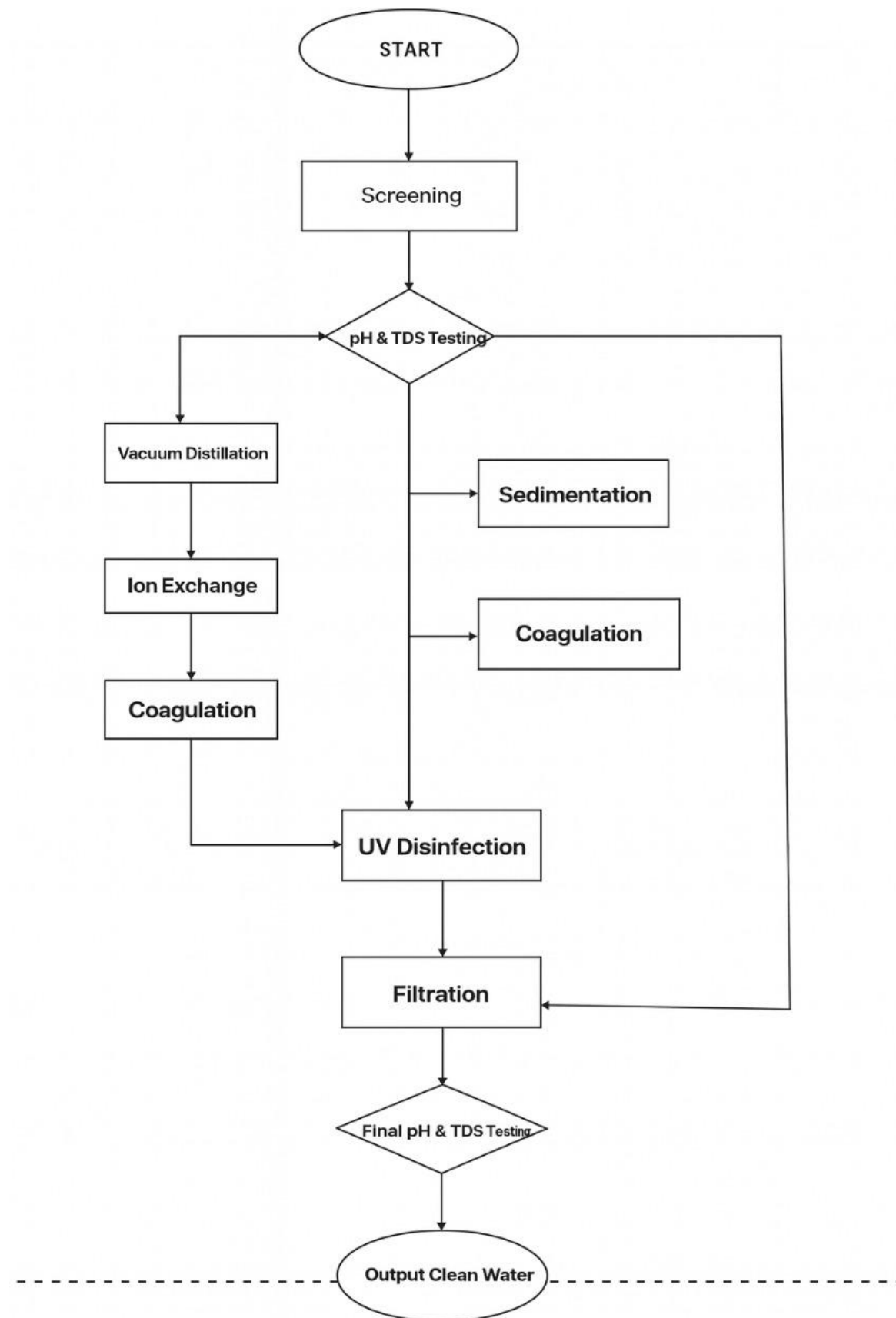
# CAD Model of Project:



# Classification of processes by water resources:



# Purification Processes:



## **System Identifies Source Automatically:--**

- Ocean/Sea Water → High TDS, high conductivity
- Pond/River Water → Medium TDS, high turbidity
- Rainwater → Low TDS, slightly acidic

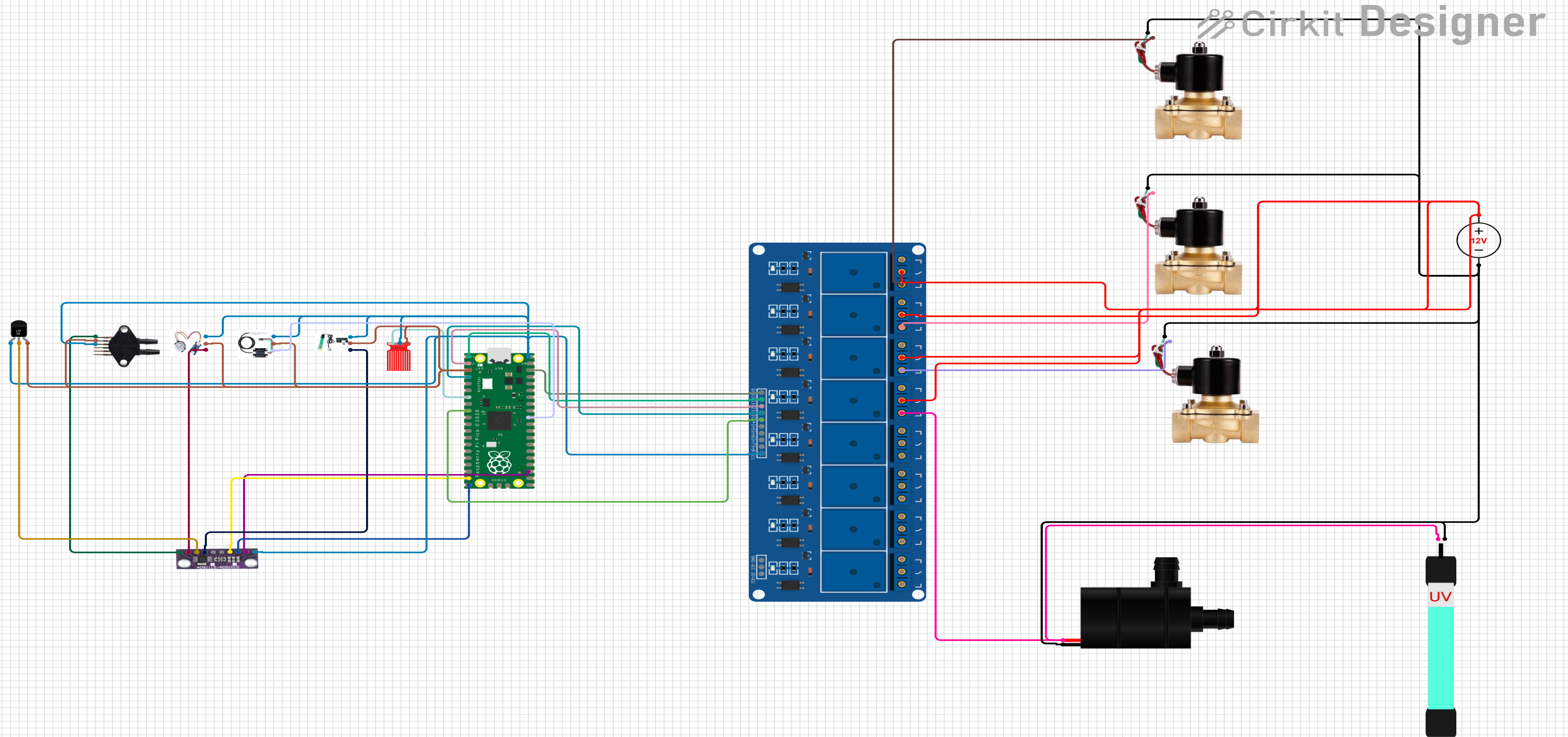
## **Classification Parameters:**

- pH Value – Measures acidity or alkalinity
- TDS (Total Dissolved Solids) – Detects mineral and salt level
- Turbidity – Indicates suspended particles
- Conductivity – Shows ion concentration

## Processes Explanation:

- **Screening** : A preliminary step that uses **mechanical screens** to remove large debris like **leaves, sticks, and plastic** from raw water.
- **Vacuum Distillation** : A **salt separation** technique from water that **lowers the boiling point** of liquids by reducing the pressure above them.
- **Ion Exchange** : A chemical method that use **anode and cathode electrodes** removes undesirable **dissolved ions** from a liquid, like water, by exchanging them for other ions with a similar charge.
- **Sedimentation** : The process where **heavier, insoluble** particles in a liquid settle to the bottom due to gravity and by adding **polyaluminium chloride(PAC)** to make it fast.
- **Coagulation** : Uses chemicals (alum, ferric chloride) called coagulants to destabilize and clump together **small suspended particles**.
- **UV Disinfection** : Uses ultraviolet light to inactivate **harmful microorganisms** like viruses, bacteria, and protozoa by damaging their DNA
- **Filtration** : A process that removes impurities from water by passing it through a filter medium, such as sand or a membrane, to trap **suspended particles, chemicals, and microorganisms** and make **hard water to soft water**.

# Circuit Connections:



# Project Timeline:

| Process/Month          | September   | October     | November    | December    | January     | February | March | April | May | June | July | August |
|------------------------|-------------|-------------|-------------|-------------|-------------|----------|-------|-------|-----|------|------|--------|
| Ideation               | <div></div> |             |             |             |             |          |       |       |     |      |      |        |
| Research               |             | <div></div> |             |             |             |          |       |       |     |      |      |        |
| Finalization           |             |             | <div></div> |             |             |          |       |       |     |      |      |        |
| CAD Model              |             |             | <div></div> |             |             |          |       |       |     |      |      |        |
| Testing/Circuit Design |             |             |             | <div></div> |             |          |       |       |     |      |      |        |
| Final Prototype        |             |             |             |             | <div></div> |          |       |       |     |      |      |        |

## Bills of Material:

| S. NO. | Items                                                                         | Unit Price | Quantity | Price |
|--------|-------------------------------------------------------------------------------|------------|----------|-------|
| 1.     | Raspberry pi module                                                           | 4000       | 1        | 4000  |
| 2.     | Analog pH sensor kit and Analog TDS sensor module and Turbidity sensor Module | 2500       | 6        | 5000  |
| 3.     | Filtration module                                                             | 4000       | 1        | 4000  |
| 4.     | Chemicals and Electrodes                                                      | 750        | -        | 750   |
| 5.     | Waterproof temperature sensor (DS18B20)                                       | 200        | 1        | 200   |
| 6.     | Pressure level Sensor                                                         | 250        | 1        | 250   |
| 7.     | Float water-level switches/solenoid valves                                    | 1500       | 3        | 4500  |
| 8.     | AC to DC convertor                                                            | 1300       | 1        | 1300  |
| 9.     | Motor and Pipeline                                                            | 3000       | 1        | 3000  |
| 10.    | Manufactured component for different Process                                  | 4000       | -        | 4000  |

Total cost = 27000 \

**THANK YOU**